Long-term effects of transcranial direct current stimulation combined with computer-assisted cognitive training in healthy older adults

See-Hyun Park[a], Jeong-Hwan Seo[a], Yun-Hee Kim[b] and Myoung-Hwan Ko[a]

The aim of the present study was to analyze the long-term effects of transcranial direct current stimulation (tDCS) of the bilateral prefrontal cortex combined with computer-assisted cognitive training on working memory and cognitive function in healthy older adults aged at least 65 years. Forty healthy older adults were enrolled and randomly assigned to two groups: anodal and sham tDCS. All participants completed 10 sessions of computer-assisted cognitive training, combined with tDCS of the bilateral prefrontal cortex. The accuracy of the verbal working memory task and performance of the digit span forward test were significantly improved after computer-assisted cognitive training combined with bifrontal anodal tDCS as compared with that after computer-assisted cognitive training combined with sham tDCS. Moreover, the effect lasts for 4 weeks in the verbal working memory task. We suggest that the tDCS-induced changes in the bilateral prefrontal excitability during computer-assisted cognitive training may have beneficial effects on age-related cognitive decrement in healthy older adults. ***NeuroReport*** **25:122–126 © 2014 Wolters Kluwer Health | Lippincott Williams & Wilkins.**

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[a]Department of Physical Medicine and Rehabilitation, Research Institute of Clinical Medicine of Chonbuk National University, Biomedical Research Institute of Chonbuk National University Hospital, Jeonbuk and [b]Department of Physical Medicine and Rehabilitation, Samsung Medical Center, Sungkyunkwan University School of Medicine, Seoul, Korea

Correspondence to Myoung-Hwan Ko, MD, PhD, Department of Physical Medicine and Rehabilitation, Research Institute of Clinical Medicine of Chonbuk National University, Biomedical Research Institute of Chonbuk National University Hospital, 664-14, 1Ga, Deokjin-dong, Jeonju-city, Jeonbuk 561-756, Korea
Tel: +82 63 250 1795; fax: +82 63 254 4145;
e-mail: mhko@jbnu.ac.kr

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Introduction

Cognitive dysfunction in older adults severely compromises their quality of life and poses social problems by increasing the healthcare burden. In this light, assessment and treatment of impaired cognitive function in older adults are becoming increasingly important.

One treatment option that is widely used for cognitive function training is computer-assisted cognitive training (CACT). CACT programs include standardized tasks and have the benefit of giving immediate feedback on task performance. Moreover, data on performance results can be continuously analyzed and compared, thereby providing beneficial effects in terms of patient outcomes as well as for clinical research. CACT has been widely used since Glisky *et al.* [1] first described it in 1986. In Korea, there have been reports that the Korean version of the CACT program significantly improved memory, as well as auditory and visual attention, in patients with brain damage [2].

Of late, an increasing number of studies are examining the effect of transcranial direct current stimulation (tDCS), a noninvasive stimulation of the cerebral cortex, on improvement of brain function. tDCS has been reported to improve spatial and temporal awareness, and attention and working memory (WM) in healthy adults, as well as in patients with brain disorders [3,4]. In particular, tDCS of the prefrontal cortex has been reported to improve WM in an increasing number of studies [5–8].

The effects of tDCS and CACT on the cognitive function in healthy adults and patients with brain disorders have been investigated separately so far; however, there is still paucity of evidence on the effect of tDCS and CACT in healthy older adults. Especially, more evidence is needed on the effect of combining the two treatments. In this light, in this study tDCS of the bilateral prefrontal cortex and CACT were concomitantly administered in the healthy elderly aged over 65 years to evaluate their long-term effect on cognitive function.

Participants and methods

Participants

This study enrolled healthy older adults aged at least 65 years without neurological abnormalities. Participants who had metallic fixtures around the cephalic area or skin lesions in the areas at which electrodes would be attached were excluded from the study. A total of 40 participants (13 men, 27 women) with a mean age of 69.7 years were enrolled. They were randomly assigned to the real stimulation and sham stimulation groups (20 participants each). Results of the single-sample χ^2-test and the independent-samples t-test showed no statistically significant differences between the real stimulation group and the sham stimulation group in terms of age,

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Table 1 General characteristics of participants

	Real group	Sham group	*P*-value
Number (male/female)	20 (7/13)	20 (6/14)	0.736
Age (years)	70.1±3.4	69.4±3.1	0.530
Education (years)	10.9±4.6	10.9±4.2	1.000
MMSE	29.3±1.4	28.8±1.5	0.330

Values are expressed as mean±SD.
MMSE, mini-mental state examination.
P-value was derived from the χ^2-test and independent-samples *t*-test.

educational background, and mini-mental state examination (Table 1). The current study was approved by the institutional review board of Chonbuk National University Hospital. Before study participation, all participants submitted a written informed consent form.

Transcranial direct current stimulation

Phoresor II Auto Model PM850 (Iomed, Salt Lake City, Utah, USA), which produces direct current, was used. Sponge electrodes of 5×5 cm (25 cm^2) were attached to the scalp. Using two stimulators, anodes were attached on the bilateral prefrontal cortex (F3 and F4 of the 10/20 international system) and cathodes were attached on the nondominant arm for bilateral prefrontal cortex stimulation [9]. The electrodes were doused in water and elastic bandages were applied to secure maximum adhesion to the respective area. Participants were randomly divided into the sham stimulation and real stimulation groups. The real stimulation group received CACT (Korean CACT program; Maxmedica Inc., Seoul, Korea) and anodal tDCS at a current intensity of 2 mA for 30 min and the sham stimulation group had an identical set-up of electrodes, but once the current intensity of 2 mA was reached, the stimulation lasted for only 30 s and the power was turned off without the participants' knowledge. The sham stimulation group was also simultaneously administered the CACT program. Different examiners were assigned to operate the transcranial direct current stimulator and to assess the cognitive function in this double-blind study.

Cognitive training program

Cognitive training was conducted using the CACT program: 30 min per session, five times a week, for 2 weeks. In total, 10 training sessions were performed along with simultaneous tDCS.

Assessment of the cognitive function

For assessment of the cognitive function, all participants underwent a total four WM assessments and computerized neuropsychological tests (CNTs): before the start of the sessions, on the day of completion of the sessions, and 7 and 28 days after the completion of the sessions.

SuperLab Pro 2.0 Software (Cedrus Corporation, San Pedro, California, USA) was used for the assessment of verbal WM. Two-back verbal WM assessment used a program showing Korean characters for 900 ms on a computer screen followed by the next character after 100 ms. The participants were required to remember the characters displayed on the computer screen and to press the spacebar on the keyboard placed in front of them in response to the same character when shown again after a different preceding character. A total of 90 characters were shown, of which 30 were 'target characters'. At each assessment, different sequences of characters were shown. Accuracy was measured by calculating the ratio of correct participant responses to 30 targets. The time taken by an individual participant to respond by pressing the spacebar after seeing a target character was also measured.

CNT was used to assess a total of 10 areas, and one examiner performed all tests. The following tests were performed: digit span test and verbal learning test, which were used for assessing verbal memory; visual span test and visual learning test, which were used for the assessment of temporal and spatial memory; auditory continuous performance test (CPT), auditory controlled CPT, visual CPT, visual controlled CPT, and the word–color test, which were used to assess attention; and the trail making test, which was used for assessing coordination function [10].

Statistical analysis

Demographic data were analyzed by the independent-samples *t*-test and the single-sample χ^2-test for categorical and continuous data, respectively. The effects of the real and sham stimulations were compared with each other using analysis of variance (ANOVA) for repeated measures. If the interaction effect (group × time) was significant, we performed repeated measures ANOVA within the real stimulation group. The statistical significance for these tests was defined as P less than 0.05.

Results

All participants underwent tDCS and CACT, verbal WM assessment, and CNT without manifesting any particular side effects.

Verbal working memory task

Comparison of results before and after tDCS between the real stimulation group and the sham stimulation group using repeated measures ANOVA showed that the accuracy of the verbal WM task had increased significantly ($P = 0.040$) and the reaction time of the verbal WM task was significantly shortened in the real stimulation group ($P = 0.050$; Table 2).

The accuracy of the verbal WM task performance was significantly increased for up to 28 days after completion of the sessions compared with that before stimulation in the real stimulation group ($P = 0.002$), and the reaction time was significantly shortened on the day of completion

Table 2 **Results of the two-back verbal working memory task and digit span forward performance**

	Group	Baseline	Day 0	Day 7	Day 28	*P*-value
Two-back verbal WM task						
Accuracy (%)	Real	48.82±10.66	57.96±11.35*	57.96±13.49*	56.94±13.13*	0.040
	Sham	50.29±11.53	51.48±13.04	52.97±13.30	54.47±14.04	
RT (ms)	Real	570.01±94.45	521.66±56.60*	530.73±70.36	531.94±73.79	0.050
	Sham	539.51±76.04	566.12±85.04	523.14±92.13	525.59±94.83	
Digit span forward (number of digits)						
	Real	4.65±1.09	5.00±1.17	5.30±0.98*	4.65±1.27	0.041
	Sham	4.70±0.08	4.95±0.83	4.80±0.83	5.00±0.73	

Values are expressed as mean±SD.
Day 0, on the day of completion of the sessions; Day 7, 7 days after completion of the sessions; Day 28, 28 days after completion of the sessions; RT, reaction time; WM, working memory.
P-value was derived from repeated measures analysis of variance for the difference between real and sham stimulation groups.
*$P<0.05$, compared with baseline.

of the sessions compared with that before stimulation ($P=0.018$; Fig. 1).

Computerized neuropsychological test

Comparison of digit span forward test results using repeated measures ANOVA showed that the real stimulation group showed a significant increase in the range of numbers that could be remembered compared with the sham stimulation group ($P=0.041$; Table 2).

However, the visual span test and visual learning test assessing temporal and spatial memory; auditory CPT, auditory controlled CPT, visual CPT, visual controlled CPT, and the word–color test assessing attention; and the trail making test assessing visual motor coordination function, showed no significant changes.

Discussion

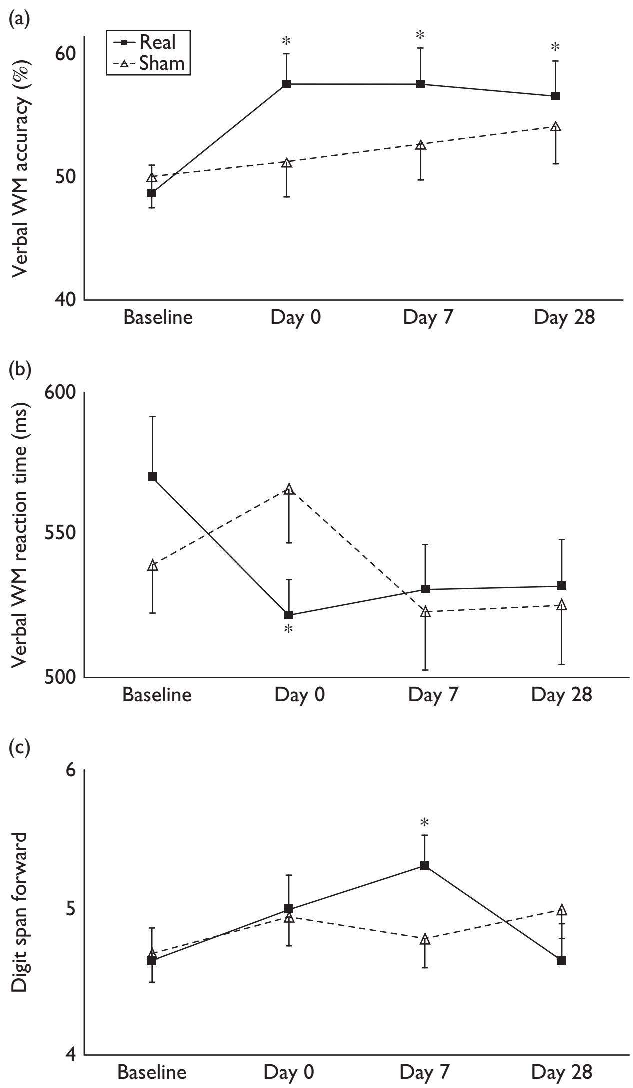
The group with simultaneous administration of the tDCS and CACT programs showed significant improvements in the verbal WM task and digit span forward test performance compared with the group in which only the CACT program was administered.

WM refers to the function of retaining information for a short period of time and is an important part of cognitive processes including long-term memory, linguistic learning and executive function, etc. [11]. In many of the previous studies, the prefrontal cortex has been reported to be responsible for WM [12]. The areas of the brain that are known to be involved in the execution of WM tasks are the bilateral medial posterior parietal cortex, bilateral premotor cortex, cingulate gyrus, bilateral frontal pole, bilateral dorsolateral prefrontal cortex, and bilateral ventrolateral prefrontal cortex [13]. It is known that execution of verbal WM tasks mainly occurs because of the activation of the left ventrolateral prefrontal cortex and the bilateral premotor cortex, the bilateral medial posterior parietal cortex, and the thalamus, whereas execution of nonverbal WM tasks mainly occurs because of the activation of the bilateral frontal pole and cingulate gyrus.

Several studies have reported that noninvasive brain stimulation can significantly improve WM in healthy adults, patients with Parkinson's disease, and stroke victims [6–8,14]. Boggio *et al.* [6] reported that a 20-min direct current stimulation of the left prefrontal cortex at an intensity of 2 mA in 18 patients with Parkinson's disease improved the verbal WM accuracy (three-back WM). Jo *et al.* [8] reported that 30 min of tDCS of the left prefrontal cortex at an intensity of 2 mA in 10 patients with cerebral infarction improved the verbal WM accuracy (two-back WM). A study involving 15 young adults conducted by Fregni *et al.* [7] demonstrated that 10 min of tDCS of the left prefrontal cortex at a current intensity of 1 mA improved the verbal WM accuracy (three-back WM). Most of the studies published until now compared changes in the cognitive function in the resting state before and after tDCS; however, tDCS combined with specialized cognitive training was not used. Although tDCS includes the term 'stimulation', in actuality this electrophysiologic technique uses low levels of electricity that are not sufficient for inducing firing of resting neurons. In other words, tDCS does not induce automatic firing of resting neurons, but it changes the neuronal resting phase thresholds through several minutes of direct current stimulation so that the degree and frequency of neuronal firing can be changed when the neurons are capable of active firing after tDCS [15]. Thus, the combination of tDCS and cognitive function training can lead to better task performance than that on administration of tDCS alone. On the basis of these results, tDCS and CACT programs were concomitantly administered in this study, and it was demonstrated that the combination of the two treatments significantly improved the cognitive function compared with administration of the CACT program alone.

Until now, studies have mainly focused on the left frontal lobe; however, in this study electrodes were attached bilaterally on the prefrontal cortex, and two stimulators were used to produce bilateral prefrontal cortex stimulation. Older adults have been reported to rely more heavily on bilateral frontal lobes to execute WM tasks, regardless of the type of stimulation [16]. Moreover, previous reports on the safety of tDCS in healthy adults were also taken into consideration in this study. Koenigs *et al.* [9]

Fig. 1



Effects of bilateral prefrontal tDCS on (a) the accuracy of the verbal WM task, (b) the reaction time of the verbal WM task, and (c) the number of digits in the digit span forward test. The improved verbal WM accuracy was maintained for 28 days after the completion of the sessions (baseline–Day 28). The reaction time of the verbal WM task was significantly shortened by tDCS between that before the start of the sessions and that on the day of completion of the sessions (baseline–Day 0). The number of digits in the digit span forward test was significantly improved by tDCS between that before the start of the sessions and that 7 days after the completion of the sessions (baseline–Day 7). Values are presented as mean±SEM. $^{*}P<0.05$. tDCS, transcranial direct current stimulation; WM, working memory.

examined healthy adults using extracephalic reference electrodes. In their study, anodes or cathodes producing a current intensity of 2.5 mA for 35 min were attached over bilateral frontal lobes, and no significant changes in terms of mood, arousal, emotional cognitions, etc. and no significant side effects were detected. The results of this study are in line with those of previous studies and no significant side effects were observed other than minimal skin discoloration on the arms on which negative reference electrodes were attached in two participants. The discoloration persisted for only a few days.

Results of this study showed that improvements were mainly achieved in verbal WM after stimulation of the bilateral prefrontal cortex; however, there were no significant improvements in performance of visual tasks. The frontal lobe plays a key role in executive functions that are related to given tasks; however, the parietal lobe plays an important role in initial recognition of a visual task. This may explain the lack of improvement in visual tasks following stimulation of the prefrontal cortex only. In addition, in the CNT there was no significant improvement in the attention task and in the trail making test assessing visual motor coordination function. Performance of these tasks requires the involvement of more areas of the brain, and therefore stimulation of the prefrontal cortex alone is insufficient to induce significant changes. New stimulation methods that have an effect on various areas of the brain need to be developed in the future.

Conclusion

The results of this study suggested that the combination of cognitive training and tDCS in healthy elderly patients can lead to improved cognitive function. In addition, this study showed that tDCS over the bilateral prefrontal cortex using extracephalic reference electrodes is safe and can contribute to improvement in cognitive function. Thus, tDCS can be expected to be used as an effective treatment method for improving the cognitive function in patients needing cognitive training.

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Conflicts of interest

There are no conflicts of interest.

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